

# Ashton E. Thomas

## EDUCATION

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| <b>University of Michigan - Rackham Graduate School</b><br><i>Major - MSE Computer Engineering, Computer Vision Concentration</i><br><i>summa cum laude, GPA - 4.00/4.00</i>                                           | 08/2025 – 05/2026<br><i>Ann Arbor, MI, USA</i> |
| <b>University of Michigan College of Engineering</b><br><i>Major - BSE Computer Science; Minor - International Engineering</i><br><i>summa cum laude, GPA - 3.8/4.0</i>                                                | 08/2022 – 05/2025<br><i>Ann Arbor, MI, USA</i> |
| <i>Relevant coursework: AR/VR, Computer Organization, Computer Vision, Cryptography, Data Analytics</i><br><i>Data Structures &amp; Algorithms, Operating Systems, Quantum Computing, Web Development, Web Systems</i> |                                                |

## EXPERIENCE

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| <b>Amazon</b><br><i>Software Development Engineer I, AWS</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 09/2024 – Present<br><i>Boston, MA &amp; New York, NY, USA</i> |
| <ul style="list-style-type: none"><li>Build developer-insights platforms at AWS for software delivery health, observability, and engineering analytics as part of a 12+ person team.</li><li>Completed an SDE internship on Alexa endpoint experiences, optimizing customer-facing device functionality with a 20+ engineer team using React Native, Kotlin, and TypeScript.</li><li>Managed Amazon company-wide employee data for 100,000s of employees as part of a 10-engineer team, including collection, processing, and executive-level inferences.</li></ul> |                                                                |
| <b>Ground Vehicle Systems Center (SEC)</b><br><i>Software Development Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 05/2024 – 07/2024<br><i>Warren, MI, USA</i>                    |
| <ul style="list-style-type: none"><li>Leveraged MagicDraw &amp; Excel to design databases for Jira tickets and hardware, leading to improved efficiency.</li><li>Created Python scripts to parse large csvs with 1000s of datapoints to update integrated networks in Jira.</li></ul>                                                                                                                                                                                                                                                                               |                                                                |
| <b>Madi Taylor Photo</b><br><i>Full Stack Developer</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 06/2022 – 07/2024<br><i>Traverse City, MI, USA</i>             |
| <ul style="list-style-type: none"><li>Developed and maintained the corporate website, crafting a cohesive user interface with HTML, CSS, and JS.</li><li>Implemented robust back-end payment solutions and form validation to streamline user transactions.</li></ul>                                                                                                                                                                                                                                                                                               |                                                                |

## RESEARCH

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| <b>Polk Lab — University of Michigan</b><br><i>Computational Neuroscience Research Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 09/2024 – 06/2025<br><i>Ann Arbor, MI, USA</i> |
| <ul style="list-style-type: none"><li>Collaborated with a PhD student to develop a machine learning model aimed at mimicking neural distinctiveness of the human brain, while also enabling the separation of background noise from spoken language.</li><li>Worked alongside Prof. Thad Polk to develop a script that processes CSV files with 1000s of columns, utilizing wildcard parsing to dynamically filter columns. The script includes a node hierarchy for efficient column interrelation and supports operations such as adding, removing, printing, and row filtration.</li></ul> |                                                |

## PROJECTS

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| <b>Geoguessr AI, Computer Vision</b>                                                                                                                                                                                                                                                                                                                                                          | 2023 – 2024 |
| <ul style="list-style-type: none"><li>Designed and implemented a modified ResNet-50 architecture for geographic location identification from images.</li><li>Fine-tuned the model with 61k images, addressing lighting and seasonal challenges whilst achieving ~90% accuracy.</li></ul>                                                                                                      |             |
| <b>Virtual Memory Pager, Operating Systems</b>                                                                                                                                                                                                                                                                                                                                                | 2025        |
| <ul style="list-style-type: none"><li>Built a virtual-memory pager managing per-process arenas, page tables, physical frames, swap-backed pages, and shared file-backed mappings.</li><li>Implemented page-fault handling, copy-on-write behavior, dirty/reference tracking, swap allocation, file-block sharing, eviction, and a clock-style page replacement policy.</li></ul>              |             |
| <b>Invariant Learning Experiments, Machine Learning</b>                                                                                                                                                                                                                                                                                                                                       | 2025        |
| <ul style="list-style-type: none"><li>Compared empirical risk minimization, invariant risk minimization, and IRM-Games under distribution shift across controlled experimental environments.</li><li>Evaluated model behavior on Colored MNIST, synthetic chain-equation data, and financial time-series settings with spurious correlations, trend regimes, and volatility shifts.</li></ul> |             |

## TECHNICAL SKILLS

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| ARM, ARCGIS, AWS, Lambda, S3, Glue, Athena, CloudWatch, Redshift, CDK, C#, C++, CI/CD, CSS, Excel, Flask, Git, HTML, Java, Javascript, Jinja, Jira, Keras, Kotlin, Latex, MagicDraw, Matlab, NumPy, Pandas, Python, PyTorch, Qiskit, React/Native, SQL, TensorFlow, Typescript, VSCode |
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